Akoizo Original Practice - Form 3 - Answer key

Division B | Codebusters | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Which word fits ABBC?

Unit: Monoalphabetic substitution and language evidence.
Worked-case data: A ciphertext word ABBC is proposed to mean MEET or TEAM. Which fits the equality pattern?

A. MEET

B. TEAM

C. TREE

D. NOON

Correct: A

2. (2 points) What forces M = 1 in the ordinary SEND + MORE example?

A. The extra leading carry

B. Spanish word frequencies

C. The letter M being common

D. Every M always means 1

Correct: A

3. (2 points) What is the key in a Caesar transformation?

Unit: Cryptography foundations and the solving workflow.
Worked-case data: A = 0 and a Caesar key of +3 are used to encode CAT. Decode FDW and verify.

A. The recovered English sentence

B. The alphabet itself

C. The paper layout

D. The shift amount

Correct: D

4. (2 points) A fractionated-Morse triple can cross what?

Unit: Baconian and fractionated Morse systems.
Worked-case data: Decode AAAAA AAAAB AABAA under the modern 26-letter Baconian convention.

A. No boundary ever

B. Only complete words

C. An original letter boundary

D. Only a page boundary

Correct: C

5. (2 points) Which check is strongest after a plausible solve?

Unit: Cryptography foundations and the solving workflow.
Worked-case data: A = 0 and a Caesar key of +3 are used to encode CAT. Decode FDW and verify.

A. Re-encrypt and compare every symbol

B. Accept any grammatical sentence

C. Ignore repeated letters

D. Count only words

Correct: A

6. (4 points) Explain “Polybius square” in your own words.

Unit: Porta, Nihilist, and checkerboard tables.
Worked-case data: Nihilist ciphertext: 65. Key coordinate: 23. Square entry: Row 4, column 2 contains R.
Separate investigation observations:
Nihilist ciphertext: 65.
Key coordinate: 23.
Square entry: Row 4, column 2 contains R.

answer: A letter table indexed by row and column. (4 points).

7. (2 points) Which priority follows the example assumptions?

Unit: Competition strategy and deliberate practice.
Worked-case data: Puzzle A offers 100 points with a 0.9 estimated solve probability in 3 minutes; B offers 200 with 0.3 probability in 6 minutes. Compare expected points per minute as a planning aid.

A. A before B

B. B necessarily before A

C. Both guarantee full credit

D. Neither can be compared

Correct: A

8. (6 points) Solve this case and explain your reasoning: Cipher: Caesar with forward encryption shift 3. Ciphertext: FDW. Alphabet: A=0 through Z=25.

reasoning: Decode by shifting backward three places.
F becomes C, D becomes A, and W becomes T.
Re-encrypt CAT to check that it gives FDW. (3 points).

conclusion: CAT (3 points).

9. (4 points) Encode C in the modern Baconian table.

Unit: Baconian and fractionated Morse systems.
Worked-case data: Decode AAAAA AAAAB AABAA under the modern 26-letter Baconian convention.

answer: AAABA: index 2 is binary 00010. (4 points).

10. (6 points) Solve this case and explain your reasoning: Rectangle: Six letters in two columns. Ciphertext pieces: CTO and ATO. Clue: The plaintext starts with CAT.

reasoning: Each column contains 6/2 = 3 letters.
Read the rows CA, TT, and OO.
Join the rows as CATTOO, which agrees with the supplied crib. (3 points).

conclusion: CATTOO (3 points).

11. (4 points) Why cannot a triple always be read as one Morse letter?

Unit: Baconian and fractionated Morse systems.
Worked-case data: Decode AAAAA AAAAB AABAA under the modern 26-letter Baconian convention.

answer: It may contain fragments from two letters or a separator, so the complete recovered stream must be parsed. (4 points).

12. (4 points) A guessed word works once but contradicts another occurrence. What should change?

answer: Revisit the guess or mapping direction; a consistent substitution must satisfy all occurrences. (4 points).

13. (2 points) Define “Polybius square” in the context of Codebusters.

answer: A letter table indexed by row and column. (2 points).

14. (4 points) Give a prevention rule for wrong key direction.

Unit: Competition strategy and deliberate practice.
Worked-case data: Puzzle A offers 100 points with a 0.9 estimated solve probability in 3 minutes; B offers 200 with 0.3 probability in 6 minutes. Compare expected points per minute as a planning aid.

answer: Write plaintext and ciphertext rows explicitly, test one known pair, then re-encrypt the finished answer. (4 points).

15. (2 points) Define “Atbash” in the context of Codebusters.

answer: A reversal of the alphabet. (2 points).

16. (4 points) Describe a complete verification.

Unit: Baconian and fractionated Morse systems.
Worked-case data: Decode AAAAA AAAAB AABAA under the modern 26-letter Baconian convention.

answer: Reconstruct intermediate groups from the proposed plaintext using the same table, separators, and padding, then compare the full encoded stream. (4 points).

17. (2 points) Define “Column height” in the context of Codebusters.

answer: Number of rows in a complete transposition rectangle. (2 points).

18. (6 points) Solve this case and explain your reasoning: A ciphertext word ABBC is proposed to mean MEET or TEAM. Which fits the equality pattern?

reasoning: ABBC has pattern 0-1-1-2.
MEET has 0-1-1-2; TEAM has 0-1-2-3.
Accept MEET only as a candidate; test A -> M, B -> E, C -> T throughout the message. (3 points).

conclusion: MEET fits the local pattern, but a full solution requires global verification. (3 points).